# MOHD SHEHZAD

**Software Development Engineer • Academic Researcher • Competitive Podiums**

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## PROFESSIONAL SUMMARY

High-achieving Computer Science undergraduate (CGPA 8.6/10) at GNDEC Ludhiana with proven dual competency in full-stack software engineering and technical communication. Creator of 4+ production web applications (React, TypeScript, Node.js, Python, PostgreSQL/Supabase), J-PAL South Asia research contributor (Columbia/Stanford/UIUC empirical study), and Grade-A certified AI/ML practitioner. Demonstrated leadership across podium finishes in competitive algorithmic debugging, 8-hour inter-college hackathons, and state-level public speaking (1st Position Declamation Champion). Seeking Software Development Engineer (SDE), Full-Stack Developer, or Associate Product/Tech roles.

## EDUCATION & ACADEMIC DISTINCTION

**Guru Nanak Dev Engineering College (GNDEC), Ludhiana** | 2023 – 2027

**Bachelor of Technology (B.Tech), Computer Science & Engineering — Current CGPA: 8.6 / 10.0**

- **Core Coursework:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, System Design Fundamentals.
- **Stage Felicitation:** Felicitated on the Main Auditorium stage on GNDEC 68th Foundation Day by Principal Dr. Sehijpal Singh for student initiatives, scholastic merit, and leadership.

**Islamia Senior Secondary School, Malerkotla** | Score: 89%

Senior Secondary (12th Grade) • Conferred with Academic Excellence & Outstanding Student Award

**Islamia Senior Secondary School, Malerkotla** | Score: 87%

Secondary Matriculation (10th Grade)

## CORE TECHNICAL COMPETENCIES

- **Programming Languages:** C, C++, Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Frontend Engineering:** React.js, Tailwind CSS, Vite, Framer Motion, HTML5 Canvas API, Component State Management, Responsive Design
- **Backend & Databases:** Node.js, Express.js, RESTful APIs, Supabase, PostgreSQL, Relational DB Modeling, JWT Auth
- **Tools & DevOps:** Git, GitHub, Linux/Bash, Postman, Vercel, Netlify, Render, Agile Development, Systems Debugging
- **AI & Machine Learning:** NumPy, Pandas, Scikit-Learn fundamentals, Generative AI developer workflows, Prompt Engineering (BE10x Certified)

## FLAGSHIP PRODUCTION PROJECTS

**1. HostelEase — Location-Based Accommodation Discovery Platform** | hostel-ease-sigma.vercel.app

**Stack: React.js, TypeScript, Tailwind CSS, Supabase (PostgreSQL), Vite**

- Engineered full-stack accommodation discovery platform for university students featuring location-based service filtering, verified property listings, and real-time inquiry workflows.
- Modeled relational PostgreSQL schema on Supabase with real-time state synchronization, maintaining sub-100ms response times and consistent UI state through reusable, type-safe React components.

**2. OmniLab Studio — Interactive Physics Simulation & Virtual Lab Workbench** | omni-physics-lab.vercel.app

**Stack: React.js, TypeScript, HTML5 Canvas API, WebGL, Tailwind CSS**

- Built in-browser physics simulation engine running 8 real-time laboratory experiments (Wave Optics, Pendulum Kinematics, Circuit Resistivity) by implementing governing physics equations in client-side runtime.
- Developed reactive parameter controls updating simulations at 60 FPS, with automated lab report generation using efficient React state management.

**3. Track Expenser — Expense Intelligence & Financial Analytics Application** | track-expenser.vercel.app

**Stack: React.js, Chart.js, Supabase, Tailwind CSS, REST APIs**

- Developed financial tracking app with user authentication, multi-category transaction management, and budget forecasting by integrating Supabase with REST API calls; visualized breakdowns with Chart.js.

**4. ZID Fitness Zone & Taskly — Management & Productivity Applications** | zid-fitness-zone.onrender.com

**Stack: React.js, Node.js/Express, Supabase, Tailwind CSS**

- Deployed full-stack role-based member workflows and attendance tracking alongside Kanban task manager with persistent drag-and-drop mechanics.

## RESEARCH EXPERIENCE & INDUSTRIAL TRAINING

**Research Contributor — "Digital Wellness in India" | J-PAL South Asia** | 2024

- Contributed to empirical field data aggregation, user metric tracking, and digital device analytics for a multi-institution research study led by researchers from Columbia University, Stanford University, and UIUC.

**Industrial Training in AI & ML using Python | ThinkNEXT Technologies (ISO 9001:2015)** | **Grade A (Outstanding)**

- Completed intensive 1-month program covering Machine Learning regression, classification, and data analytics pipelines using Python, NumPy, and Pandas (Cert ID: 939895).

## LEADERSHIP, COMPETITIVE PODIUMS & HONORS

- **3rd Prize (Podium Finish), "Debugging using C":** Tech Disha 2024, Arya College (Outperformed regional competitors in C code debugging, memory resolution, and algorithmic problem-solving).
- **First Position (Champion), Grand Declamation Contest:** Conversa English Academy & Azad Foundation (Won 1st Position for persuasive rhetoric and oratory on "Plant Tree, Save Earth").
- **2nd Position, Grand Declamation Contest "WE ARE ONE":** Awake Foundation & Radiance Foundation (Awarded 2nd Position competing across 51+ schools and colleges with 510+ participants).
- **Contender, THINK-A-THON 2026:** 8-Hour Inter-College Hackathon by ITian Club, GNDEC ("Where Ideas Ignite Impact").
- **2nd Position (Podium Finish), Online Case Study Making:** Center for Multi-Faceted Learning (CML), GNDEC (Commended for creative problem formulation and presentation delivery).
- **Stage Felicitation Honor:** GNDEC 68th Foundation Day (Felicitated by Principal Dr. Sehijpal Singh on the Main Auditorium stage).
- **State Honours Ceremony Memento:** All India Lohar Trust (Regd.) Punjab (Honored at the state ceremony for community leadership and scholastic distinction).
- **University Athlete:** 64th Annual Athletics Meet, GNDEC (Competed in 100m Sprint & Long Jump representing CSE Department).

## TEACHING & MENTORSHIP (LEADERSHIP & COMMUNICATION)

**Academic Tutor & STEM Mentor | Independent Tutoring | Malerkotla, Punjab** | 2024 – Present (1 Year)

- Mentored 15+ students across grades 5 through 10 in Mathematics, Science (Physics/Chemistry), and Computer Science fundamentals, improving term grades by 15–20% through structured diagnostic quizzes and personalized pedagogy.

## VERIFIED INDUSTRY CERTIFICATIONS

- **Python Foundation Certification (Capstone + Parts 1 & 2):** Infosys Springboard (Accredited Python curriculum covering OOP, data structures & capstone assessment)
- **WEB FORGE 2026 Web Engineering Challenge:** GNDEC Data Science Club (Ref: CSE/DSC/4681 • Evaluated by departmental faculty)
- **AI Tools & Modern Engineering Acceleration Workshop:** BE10x (Generative AI developer workflows, prompt engineering, and developer automation)
- **International IoT Technology & Engineering Webinar:** American Association of Engineers of Indian Origin (AAEIO, USA) & GNDEC DSC